Ha (Haden) H. Nguyen

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Professional Summary

M.S. candidate in Engineering Sciences with experience in computational mechanics, elastodynamics, finite element methods, scientific computing, simulation-driven engineering, and machine learning. Background includes high-performance FEM development, wave-propagation modeling, aerospace structural design, and data-driven surrogate modeling. Skilled in Python, Mathematica, Abaqus, FEMAP Nastran, SolidWorks, and CATIA, with strong expertise in mathematical modeling, computational implementation, technical communication, presentations, and scientific writing.

Education

University of Colorado Boulder — Aug 2023 – May 2026 (expected)
M.S. in Civil Engineering, Engineering Sciences — *Boulder, CO*

- Specialization: Computational mechanics and elastodynamics | GPA: 3.69/4.00
- Relevant coursework: Nonlinear FEM, Numerical Analysis II, Material Constitutive Modeling, Complex Variables, Continuum Mechanics.

Hanoi University of Science and Technology — Aug 2017 – Sep 2022
Engineer in Mechanical Engineering, Manufacturing Technology — *Hanoi, Vietnam*

- Thesis: *Dynamical behavior of complex plate structures under external impacts and environmental loading*

Technical Skills

Programming: Python, Mathematica, MATLAB, C++, Fortran
Simulation/FEA: Abaqus, FEMAP Nastran, ANSYS, HyperMesh
CAD: Inventor, SolidWorks, CATIA, AutoCAD
ML / Scientific Computing: Numba, Fourier Neural Operators (FNOs), PINNs, HPC-ready workflows
Documentation: LaTeX/Overleaf, reproducible technical reporting

Experience

Graduate Research Assistant — Aug 2023 – Present
University of Colorado Boulder — *Boulder, CO*

- Analyzed Green's function solutions for elastodynamic half-spaces and layered media to study wave reflection, refraction, mode conversion, and arrival behavior.
- Developed Mathematica solvers and HPC-ready computational workflows for transient wave-propagation modeling.
- Extended Adaptive-Gradient FEM codebase.
- Built Fourier Neural Operator surrogates and ML-ready simulation pipelines for fast elastodynamic wavefield prediction.
- Implemented an uncertainty-aware, data-driven equation discovery workflow for computational mechanics; manuscript in preparation.

Research Intern — Jun 2025 – Aug 2025
Los Alamos National Laboratory — *Los Alamos, NM*

- Implemented and validated a Numba-accelerated Python FEM solver for static and dynamic analysis using a novel Adaptive-Gradient finite element formulation.
- Developed benchmark, patch, and verification tests to assess numerical accuracy and robustness.
- Demonstrated improved accuracy through analytical validation and comparative studies against conven-

tional finite elements.

Aerospace Structural Engineer Oct 2022 – Jul 2023
Viettel Aerospace Institute *Hanoi, Vietnam*

- Designed aerospace structural parts and assemblies in Inventor, SolidWorks, and CATIA, and generated manufacturing-ready technical drawings for fabrication and assembly.
- Developed and analyzed finite element models in FEMAP Nastran and Abaqus to evaluate stress, deformation, and dynamic response under operational and qualification loading conditions.
- Executed deformation and vibration tests, including random vibration testing, to validate structural performance and support verification and qualification efforts.

Undergraduate Researcher Jan 2020 – Aug 2023
Hanoi University of Science and Technology *Hanoi, Vietnam*

- Developed extended plate/shell theory models and MATLAB solvers for vibration and chaotic-response analysis of structures with geometric complexity, cracks, variable thickness, and advanced composite materials.
- Contributed to multiple peer-reviewed publications in computational mechanics, structural dynamics, and machine-learning-enabled modeling.

Engineering Intern Mar 2022 – May 2022
Thanh Anh Co., Ltd *Hanoi, Vietnam*

- Designed manufacturing processes, jigs, and fixtures and generated CNC programs to support batch production of mechanical parts.

Teaching Experience

Teaching Assistant Fall 2023 – Present
University of Colorado Boulder *Boulder, CO*

- Supported mechanics and civil engineering courses through office hours, grading, lab instruction, and exam support for undergraduate students.
- Assisted with Dynamics, Mechanics of Materials, Advanced Mechanics of Materials, and FE exam preparation courses.

Publications

11 peer-reviewed publications, with additional manuscripts under review or in preparation, in journals including *AIAA Journal*, *Aerospace Science and Technology*, *Thin-Walled Structures*, *Engineering Structures*, *Ocean Engineering*, and *Composite Structures*.

Selected Publications

- Pak, R. Y. S., **Nguyen, H. H.** *Improved FE Treatment of Classical Rigid Punch Problems on a Layer Using Adaptive-Gradient Elements*. In preparation.
- Pak, R. Y. S., **Nguyen, H. H.** *Green's Function Solution for Transient Wave Propagation in a Homogeneous Half-Space*. In preparation.
- **Nguyen, H. H.** et al. (2023). *Dynamical and Chaotic Analyses of Single-Variable-Edge Cylindrical Panels Made of a Sandwich Auxetic Honeycomb Core Layer in a Thermal Environment. Thin-Walled Structures.*
- **Nguyen, H. H.** et al. (2022). *Research on Vibrational Characteristics of Nanocomposite Double-Variable-Edge Plates Immersed in Liquid Under the Effect of Explosive Loads. Ocean Engineering.*

Awards

- Dean's Fellowship for Excellence Award, University of Colorado Boulder ($5,000)
- Encouragement Learning Scholarship, Hanoi University of Science and Technology